Journal of Wind Engineering and Industrial Aerodynamics, 41-44 (1992) 1565–1576
Elsevier

Texas Tech field experiments for wind loads part 1: building and pressure measuring system

Marc L. Levitan[a] and Kishor C. Mehta[b]

[a]Managing Director, Wind Engineering Research Field Laboratory, Department of Civil Engineering, Texas Tech University, Lubbock, Texas, 79409, USA

[b]Professor of Civil Engineering, Texas Tech University, Lubbock, Texas, 79409, USA

Abstract

An important research facility has been constructed at Texas Tech University to study wind effects on low buildings in the field. This laboratory features many capabilities not found in past field experiments, and presents tremendous new learning opportunities. The facility is being used to pursue research in wind loads on building surfaces, internal pressures, performance of roofing in wind environment, and ventilation and exhaust studies around the building. The paper includes sufficient detail on the test building and pressure measuring system to permit researchers in physical and analytical modeling to make appropriate comparisons with the field measured data.

1. INTRODUCTION

In recent years, there have been several full-scale experiments to determine wind-induced pressures on low buildings. An important goal in each of these experiments was to obtain reliable field-measured data. This field data represents the real life wind loads on buildings and structures and provides checks on wind tunnel results as well as newly developing computational fluid dynamics technology.

In an attempt to obtain reliable wind load data, researchers at Texas Tech University have constructed a permanent laboratory to measure wind effects on structures in the field. This facility combines the best features of past field experiments with advanced instrumentation and computer technology, providing capabilities not previously available.

Descriptions of the physical facility and preliminary data have been published previously [1,2]. Since then considerable upgrading of the instrumentation and data acquisition system has been achieved. This paper and its companion paper describe the test facility in detail, as well as present information on the characteristics of the testing systems and surrounding environment. Part I describes the test building, pressure measuring system, and data acquisition system. Details of the meteorological instrumentation systems and wind parameters present at the site are given in Part II [3]. It is intended that these two papers provide the necessary background information for those researchers wishing to use data collected at the Texas Tech field site for model studies.

2. FIELD FACILITY

The field laboratory is located on Texas Tech University land in Lubbock, Texas. It consists of a test building and a meteorological tower (see Fig. 1). The test building, pressure measuring and data acquisition systems are described in this paper. Details of the meteorological instrumentation and surrounding terrain are given in Part II.

The test building is a 30 x 45 x 13 ft (9.1 x 13.7 x 4.0 m) prefabricated metal building. The building is not anchored directly to the foundation, but rather to a rigid frame undercarriage. Wheels at the corners of the undercarriage ride on a circular steel rail embedded in the concrete slab. Hydraulic jacks at each corner are used to raise the building 3 in. (7.6 cm), allowing it to be rotated using a pair of electric motors. This system allows the building to be rotated 360°, thus providing positive control over wind angle of attack. Once rotated to the desired position, the building is lowered and secured to anchor bolts embedded in the slab. Anchor bolts are located at 15° intervals around the circle.

Centered within the outer, rotatable test building stands a fixed, concrete block building housing the data acquisition system. All signal cables and tubing for the test building enter the block building through a small circular hole in the center of the block building roof slab. This arrangement alleviates the need to disconnect and reconnect the cabling and tubing when rotating the building.

Building position is defined as the angle between the longitudinal axis of the building and true north (see Fig. 2). Wind angle of attack is the wind azimuth minus the building position. Since the terrain is essentially uniform in all directions (except to the southeast, see Part II), records are typically grouped and analyzed by angle of attack rather than wind azimuth.

More than 100 pressure taps have been installed on the test building surfaces. Taps are 5/16-in. (7.9 mm) inside diameter and 2 in. (5.1 cm) long. The corrugated metal walls and roof of the test building are covered with a layer of flatstock to give a clean, smooth surface. Taps are mounted flush with the outside building surface.

A tap numbering scheme based on tap coordinates is used. Each tap is assigned a 5-digit number, *sxxyy*. The *s* indicates which building surface the tap is located on, walls (1-4) or roof (5). The building north wall is defined as the short wall with the door and numbered 1. The other walls are numbered in a clockwise fashion. The two pairs of digits *xx* and *yy* represent the nominal coordinates (in feet) of the tap on the given surface. Figure 3 shows the surface number and origin of coordinates for each of the five building surfaces. For example, Tap 31407 indicates a tap on the building south wall, nominally 14 ft from the left edge and 07 ft high on the wall (see Fig. 3). Exact coordinates of the taps so far installed are given in Table 1.

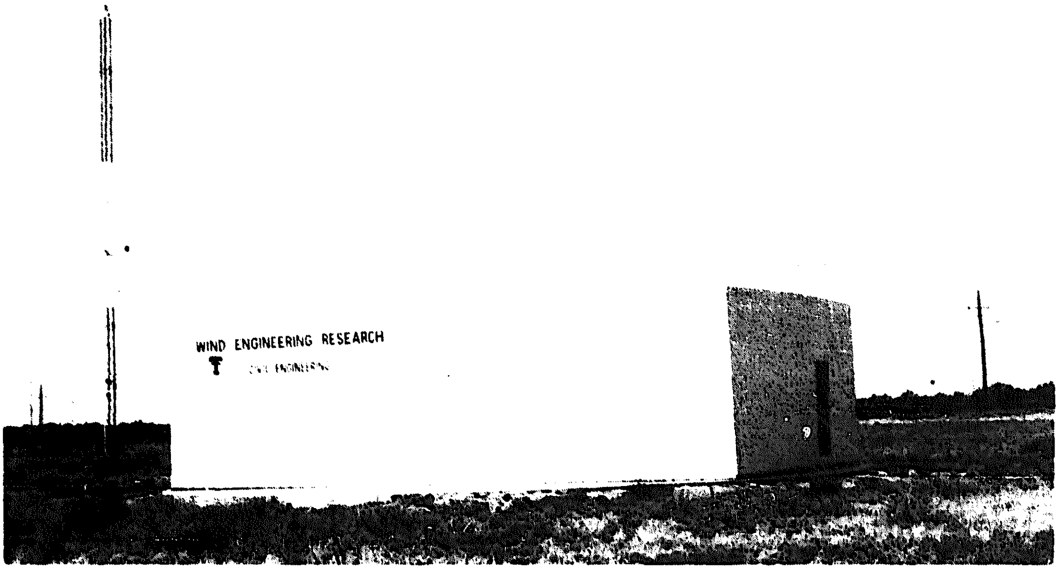

Figure 1. Test building and meteorological tower.

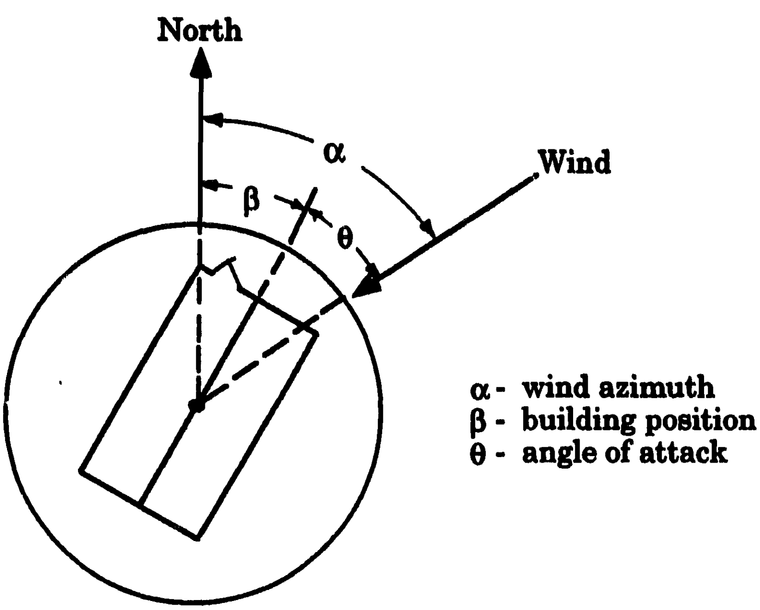

Figure 2. Wind azimuth and angle of attack.

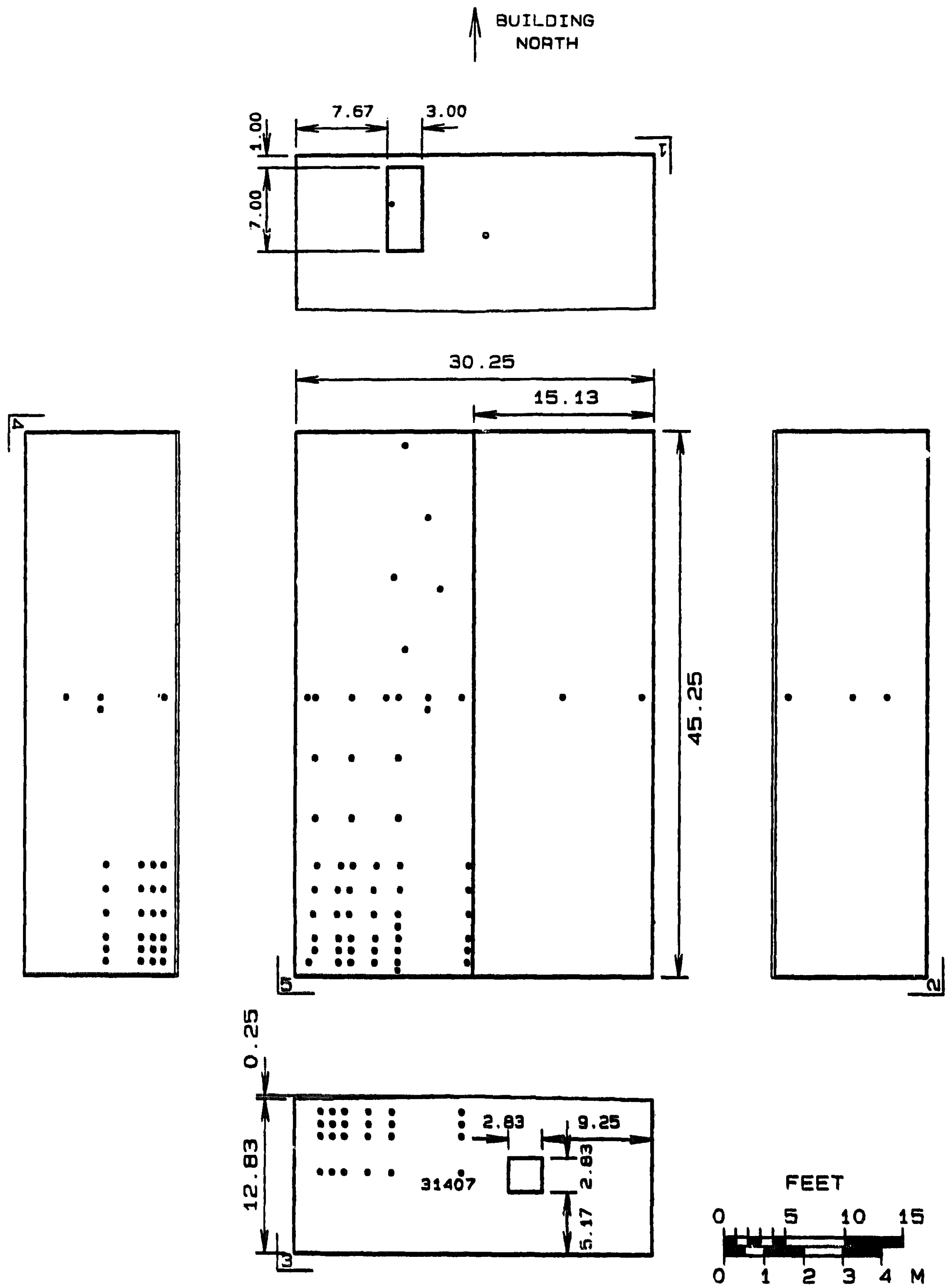

Figure 3. **Building dimensions and tap locations.**

Table 1 Pressure Tap Locations

Tap No.	x feet	x (m)	y feet	y (m)
11407[c]	14.17	(4.32)	6.75	(2.06)
22304[a]	23.08	(7.03)	3.5	(1.07)
22306[a,b,c]	23.08	(7.03)	6.42	(1.96)
22312[a,c]	23.08	(7.03)	11.83	(3.61)
30207	2.17	(0.66)	6.75	(2.06)
30210	2.17	(0.66)	9.75	(2.97)
30211	2.17	(0.66)	10.75	(3.28)
30212	2.17	(0.66)	11.75	(3.58)
30307	3.17	(0.97)	6.75	(2.06)
30310	3.17	(0.97)	9.75	(2.97)
30311	3.17	(0.97)	10.75	(3.28)
30407	4.17	(1.27)	6.75	(2.06)
30410	4.17	(1.27)	9.75	(2.97)
30411	4.17	(1.27)	10.75	(3.28)
30412	4.17	(1.27)	11.75	(3.58)
30607	6.17	(1.88)	6.75	(2.06)
30610	6.17	(1.88)	9.75	(2.97)
30611	6.17	(1.88)	10.75	(3.28)
30612	6.17	(1.88)	11.75	(3.58)
30807	8.17	(2.49)	6.75	(2.06)
30810	8.17	(2.49)	9.75	(2.97)
30811	8.17	(2.49)	10.75	(3.28)
30812	8.17	(2.49)	11.75	(3.58)
31407[c]	14.17	(4.32)	6.75	(2.06)
31410	14.17	(4.32)	9.75	(2.97)
31411	14.17	(4.32)	10.75	(3.28)
31412	14.17	(4.32)	11.75	(3.58)
42204[a]	22.17	(6.76)	3.5	(1.07)
42206[a,b,c]	22.17	(6.76)	6.42	(1.96)
42212[a,c]	22.17	(6.76)	11.83	(3.61)
42306[c]	23.17	(7.06)	6.42	(1.96)
43607	36.08	(11.00)	6.92	(2.11)
43610	36.08	(11.00)	9.92	(3.02)
43611	36.08	(11.00)	10.92	(3.33)
43612	36.08	(11.00)	11.83	(3.61)
43807	38.08	(11.61)	6.92	(2.11)
43810	38.08	(11.61)	9.92	(3.02)
43811	38.08	(11.61)	10.92	(3.33)
43812	38.08	(11.61)	11.83	(3.61)
44007	40.08	(12.22)	6.92	(2.11)
44010	40.08	(12.22)	9.92	(3.02)
44011	40.08	(12.22)	10.91	(3.33)
44012	40.08	(12.22)	11.83	(3.61)
44207	42.08	(12.83)	6.92	(2.11)
44210	42.08	(12.83)	9.92	(3.02)
44211	42.08	(12.83)	10.91	(3.33)
44212	42.08	(12.83)	11.83	(3.61)
44307	43.08	(13.13)	6.92	(2.11)
44310	43.08	(13.13)	9.92	(3.02)
44311	43.08	(13.13)	10.92	(3.33)
44312	43.08	(13.13)	11.83	(3.61)
44407	44.08	(13.43)	6.92	(2.11)
44410	44.08	(13.43)	9.92	(3.02)
44411	44.08	(13.43)	10.92	(3.33)
44412	44.08	(13.43)	11.83	(3.61)
50101[b,c]	1.17	(0.36)	1.17	(0.36)
50123[b,c]	1.00	(0.30)	23.17	(7.06)
50202	1.67	(0.51)	2.17	(0.66)
50203	1.67	(0.51)	3.17	(0.97)
50205[b,c]	1.5	(0.46)	5.17	(1.58)
50207	1.67	(0.51)	7.17	(2.18)
50209[b,c]	1.83	(0.56)	9.17	(2.79)
50213[c]	1.83	(0.56)	13.17	(4.01)
50218[c]	1.83	(0.56)	18.17	(5.54)
50223[c]	1.67	(0.51)	23.17	(7.06)
50401	3.67	(1.12)	1.17	(0.36)
50402	3.67	(1.12)	2.17	(0.66)
50403	3.67	(1.12)	3.17	(0.97)
50405	3.5	(1.07)	5.17	(1.58)
50407	3.58	(1.09)	7.17	(2.18)
50409	3.83	(1.17)	9.17	(2.79)
50501[b,c]	4.67	(1.42)	1.17	(0.36)
50502	4.67	(1.42)	2.17	(0.66)
50503	4.67	(1.42)	3.17	(0.97)
50505[b,c]	4.5	(1.37)	5.17	(1.58)
50507	4.58	(1.40)	7.17	(2.18)
50509[b,c]	4.83	(1.47)	9.17	(2.79)
50513[c]	4.67	(1.42)	13.17	(4.01)
50518[c]	4.67	(1.42)	18.17	(5.54)
50523[c]	4.67	(1.42)	23.17	(7.06)
50701	6.67	(2.03)	1.17	(0.36)
50702	6.67	(2.03)	2.17	(0.66)
50703	6.75	(2.06)	3.17	(0.97)
50705	6.58	(2.01)	5.17	(1.58)
50707	6.58	(2.01)	7.17	(2.18)
50709	6.83	(2.08)	9.17	(2.79)
50823[a,c]	7.58	(2.31)	23.17	(7.06)
50833[c]	8.25	(2.51)	33.17	(10.11)
50900[c]	8.67	(2.64)	0.50	(0.15)
50901[b,c]	8.67	(2.64)	1.17	(0.36)
50902[c]	8.67	(2.64)	2.17	(0.66)
50903[c]	8.75	(2.67)	3.17	(0.97)
50904[c]	8.67	(2.64)	4.17	(1.27)
50905[b,c]	8.67	(2.64)	5.17	(1.58)
50907[c]	8.83	(2.69)	7.17	(2.18)
50909[b,c]	8.83	(2.69)	9.17	(2.79)
50913[c]	8.67	(2.64)	13.17	(4.01)
50918[c]	8.87	(2.64)	18.17	(5.54)
50923[c]	8.67	(2.64)	23.17	(7.06)
50927[c]	9.17	(2.79)	27.17	(8.28)
50944[c]	9.17	(2.79)	44.17	(13.46)
51123[c]	11.17	(3.40)	23.17	(7.06)
51138[c]	11.17	(3.40)	38.17	(11.63)
51232[c]	12.25	(3.73)	32.17	(9.80)
51423[a,c]	14.08	(4.29)	23.17	(7.06)
51501	14.67	(4.47)	1.17	(0.36)
51502	14.67	(4.47)	2.17	(0.66)
51503	14.75	(4.50)	3.17	(0.97)
51505	14.75	(4.50)	5.17	(1.58)
51507	14.75	(4.50)	7.17	(2.18)
51509	14.75	(4.50)	9.17	(2.79)
52323[a,c]	22.58	(6.88)	23.17	(7.06)
52923[a,c]	29.25	(8.91)	23.17	(7.06)

[a]Mode M01 Transducer Locations
[b]Mode M04 Transducer Locations
[c]Mode M15 Transducer Locations

3. INSTRUMENTATION

Differential pressure transducers are used to measure surface and internal pressures. Forty-seven transducers of two different types are available for use. Additional instrumentation available includes displacement transducers and load cells. Sensors on the door and window indicate whether they are open or closed, important when analyzing internal pressure data.

Seventeen of the differential pressure transducers are supplied by Validyne Engineering Corp. (Model DP103). Of these, 13 have a full-scale (FS) range of ±0.2 psi (1.38 kPa) and four have a range of ±0.32 psi (2.21 kPa) FS. Rated accuracy for these transducers (including effects of linearity, hysteresis and repeatability) is ±0.25% FS. These transducers are very rugged and versatile, but also very expensive. Transducer housings and diaphragms are made of stainless steel and can handle almost any type of fluid. Replaceable diaphragms allow the same transducer to be used for many different pressure ranges. The transducer is a variable reluctance type. Diaphragm displacement causes a change in the magnetic field, which is converted to a voltage output linearly proportional to the applied differential pressure. Signal conditioning is provided by Validyne Engineering Model CD19A Carrier Demodulators. These units have built-in switch selectable low-pass filters (1, 10, or 50 Hz cutoff).

The other 30 transducers are OMEGA Engineering Inc. Model PX 163-005 BD 5V (manufactured by MicroSwitch), with a full-scale range of ±0.18 psi (1.24 kPa). Rated linearity for these transducers is ±0.5% FS max, and rated hysteresis and repeatability are ±0.25% FS. These transducers are solid state piezoresistive type with a fixed silicon diaphragm. Diaphragm strain is converted to an analog output voltage proportional to the applied differential pressure. Signal conditioning for these transducers was custom built. The OMEGA transducers are not as rugged or versatile as those from Validyne, but they cost only about one fourth as much.

4. PRESSURE MEASURING SYSTEM

The transducers are mounted on boards along with electrical solenoid valves (used for on-line calibration). The boards are mounted on the inside surfaces of the test building next to the pressure taps, and can easily be moved from one tap location to another. For wall taps, a 6-in. (15 cm) length of 3/8-in. (9.5 mm) inside diameter (id) flexible plastic tube connects the tap to a three-way solenoid valve (Peter Paul Electronics Co. Model 26KK9ZCM). A 10-in. (25 cm) length of 3/16-in. (4.76 mm) id tube connects the solenoid valve to the active side of the transducer. For roof taps, a 6-in. (15 cm) length of 3/8-in. id tubing extends down from the tap and connects to a tee. A 4-in. (10 cm) tube extends horizontally from the tee to the solenoid. A 12-in. (30.5 cm) length of tube extends down from the tee and is capped at the end, which acts as a reservoir to collect rainwater.

Ambient atmospheric pressure is used as reference pressure for the transducers. It is obtained from a box below ground, which has a smooth lid with a small hole (0.5-in. [1.27 mm] id). The reference pressure box is located

75 ft (23 m) west of the center of the test building. The building and the box are located far enough apart so that the building has little effect on static pressure at the reference pressure box [4]. An 8-in. (20.3 cm) diameter pipe transmits the ambient pressure to the data acquisition building. From there, 3/16-in. (4.76 mm) id tubing connects the reference pressure to each transducer. Restrictor tubing of 1/32-in. (0.79 mm) id and 40-in. (1.02 m) length is placed in the reference pressure lines to provide a time constant of about 13 seconds. This time constant is long enough to damp out high frequency noise but short enough to let through changes in atmospheric pressure.

Zero calibration of all transducers is automatically checked before and after each data run. A computer controlled solenoid valve installed on each transducer mounting board shuts off pressure from the tap and connects calibrating pressure to the active side of the transducer. Instrument zero drift is checked by supplying reference pressure to both sides of the transducer.

Dynamic calibration of the active pressure tubing system for wall-mounted and roof-mounted pressure taps has been performed. Frequency response for roof and wall tap setups for both types of transducers is shown in Fig. 4. Gain for all four setups is close to unity to about 20 Hz. Phase response of OMEGA transducer setups is extremely flat up to about 45 Hz. Validyne transducer setups show a small phase lag of 5° at 10 Hz, which increases linearly until 40 Hz. Preliminary analysis of field-measured pressure data seems to indicate that there is usually no significant energy in the pressure fluctuations above 5 Hz, meaning that the frequency response of the pressure transmission and measuring system is more than adequate.

5. DATA ACQUISITION SYSTEM

A 20 MHz 80386-based PC (with 8 MB RAM and math coprocessor) is used for data acquisition. A MetraByte Corp. DAS-8 analog-to-digital (A/D) conversion board is used to capture the incoming signals. The DAS-8 is an 8 channel, medium-speed 12-bit A/D board. Three MetraByte compatible CIO-MUX32 multiplexors from Computer Boards Inc. are used to expand input capacity to 96 channels.

LabTech Notebook software from Laboratory Technologies Corp. running in a DOS environment is used to drive the A/D board. This software runs within a custom shell written in Microsoft QuickBASIC. Incoming data is streamed directly to the hard drive during the data acquisition run. Disk caching software Super PC-Kwik from Multisoft Corp. is used to improve throughput to the disk. With this system, sustained sampling rates in excess of 2000 Hz (total for all channels) can be achieved.

At 2000 samples per second, a 15-minute long data acquisition run generates 1.8 million data points. Even using binary integer files, this corresponds to nearly 4 MB of disk storage required for each record. This can and does result in 50+ MB of data being collected on a single windy day. To handle this enormous amount of data, a 600 MB erasable optical cartridge drive is used. Immediately following each data acquisition run, the data is copied from the internal hard drive to the external optical drive.

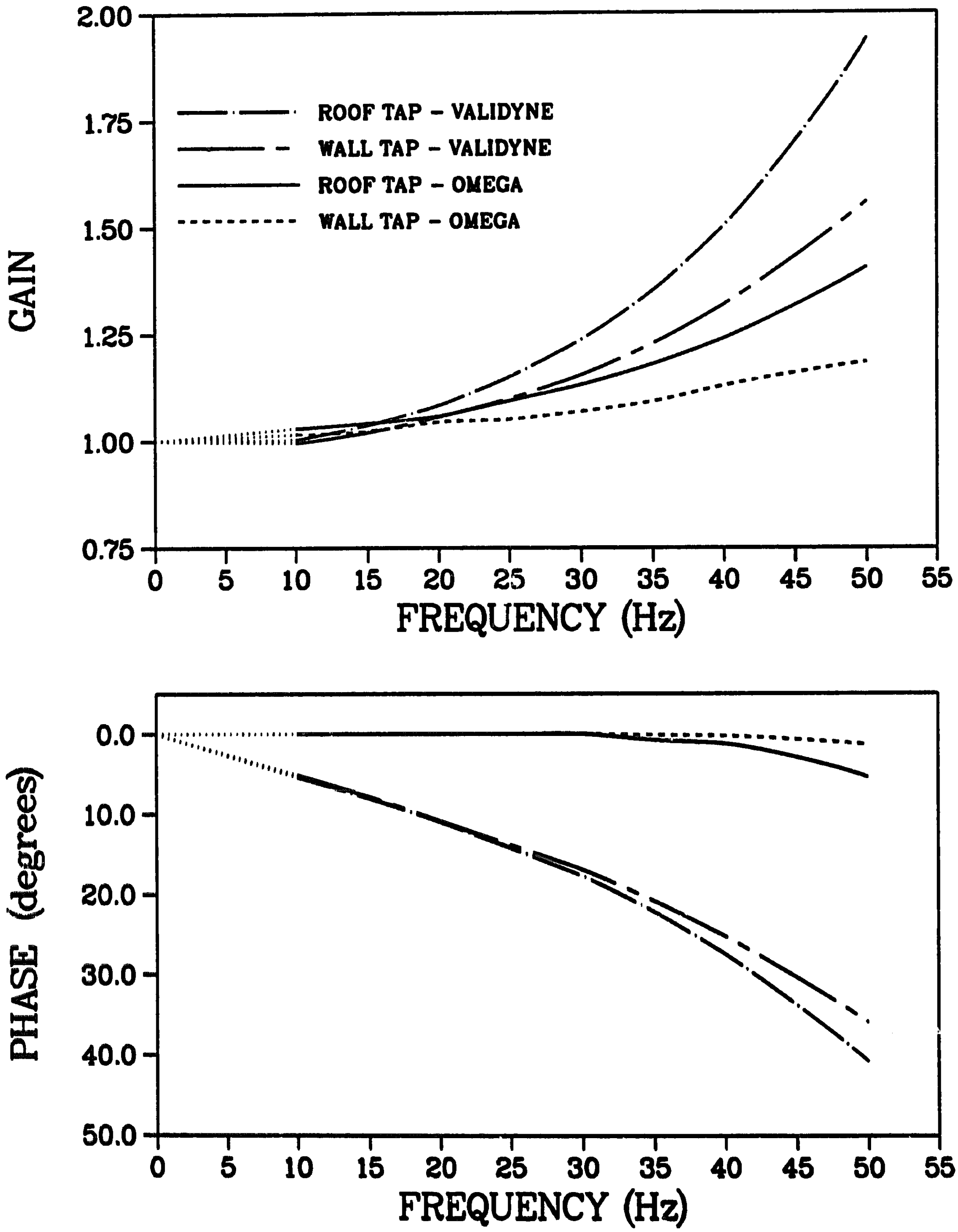

Figure 4. Frequency response of pressure transducer setups.

The data acquisition system operates continuously, monitoring the wind speed and triggering automatically when the one-minute mean speed (at the building roof height) exceeds 20 mph (9 m/s). Once triggered, the computer first performs a short (20 sec) pretest calibration run for transducer zero drift, automatically turning appropriate solenoid valves on and off. Next, the main acquisition program is run (typically 15-minute duration, depending on the specific acquisition mode selected). Once finished, a post-test zero calibration run is performed and then the triggering program is restarted.

In an extreme wind event, it is likely that there may be interruption of the regular power supply. Lubbock has experienced several partial blackouts during windstorms and thunderstorms in the past decade. To protect against this possibility, the data acquisition system is protected by two uninterruptible battery backup power supplies (Tripp Lite Model BC 1200). These units switch on immediately upon loss of power, and are powerful enough to run the data acquisition system and all instrumentation for approximately one hour.

Data collected at the field site is stored on optical cartridges and transported to the main campus for analysis. The optical cartridges are also used for permanent data storage and archival copies. Although the price per MB for optical storage media is currently about 50% greater than that for magnetic tape, it offers several distinct advantages. It is a much less volatile, more reliable media. It also has a shelf life of 10+ years, as opposed to the 2-4 years typical for magnetic media.

6. DATA VALIDATION SYSTEM

Certainly the single most important task for Texas Tech researchers is that of data validation and quality assurance. Due to the special nature of this project, it is absolutely imperative that the data be of unquestionable quality. Toward that end, a significant amount of the available time and resources go into a quality assurance program. This program has three major components: a daily check of the field laboratory, frequently scheduled instrument calibrations and maintenance, and timely preliminary analysis of the data collected.

The daily check consists of one of the researchers making a trip to the field site every day and performing an inspection of the major systems. A checklist of about 20 items includes: visual inspection of all instruments, particularly those outside the test building (i.e., meteorological instruments), scan of all instrument channels for "reasonableness", and check of the computerized data acquisition system. Anything out of the ordinary is noted in the daily log. This check normally takes about 30 minutes unless problems are encountered.

The second major item in the quality assurance program involves regular calibration and maintenance of all instrumentation. Pressure transducer calibration is checked every time a data run is triggered, and they are recalibrated weekly. All anemometry is wind tunnel tested and calibrated at least three times per year. Anemometer bearings are replaced at least once per year. Calibration of other meteorological instrumentation is checked at least once per week. The data acquisition system calibration is checked weekly, and A/D board and multiplexor noise level is recorded during each data run.

Finally, timely analysis of the data acquired can also point to problems with the instrumentation and data acquisition system. For example, analysis of transducer calibration runs can indicate problems with transducers or signal conditioning. Wind velocity profile can show if one of the anemometers is running slow. Analysis of channels set up to monitor electronic noise can indicate A/D or multiplexor problems.

7. DATA ANALYSIS SYSTEM

Once data is brought in from the field site, it passes through several standard analysis programs. A preprocessor converts the raw A/D integer counts into equivalent voltages and then into engineering units. Pressure transducer data is automatically corrected using calibration data taken before and after each data run. Gill UVW anemometers are corrected for non-cosine response using the manufacturers wind-tunnel-measured calibrations.

The preprocessor also computes summary statistics for each instrument in the record, and produces time history plots. At this point, an analyst thoroughly reviews each instrument in the record, examining the statistics and time histories as well as the instrument calibration and daily check logs. Validated instruments are marked and then added to the permanent database.

One of the most important statistics generated is the stationarity check of wind speed and direction data. The procedure for testing the stationarity of a single time history is as follows. First, the time history is divided into equal time intervals and the mean and variance for each interval are computed. The sequences of interval means and variances are then tested for trends using both the run test and the reverse arrangements test as described in [5]. These tests are performed at 0.05 level of significance. If any of the four tests indicate a trend, then the time history is labeled nonstationary. Wind speed and wind direction time histories of 15-minute duration are divided into 18 intervals. The procedure for choosing the number of time intervals is given in [6]. If the wind speed and direction measured at building roof height are both found to be stationary for a given record, then that record is considered stationary.

Most data analysis is performed on 80386- and 80486-based personal computers running under DOS. Preprocessor and database software was custom written in Microsoft QuickBASIC and Borland C++. Exploratory data analysis is performed using DADiSP (Data Analysis and Digital Signal Processing) software from DSP Development Corp. and CSS:STATISTICA from StatSoft. Repeated computationally intensive procedures such as correlations and spectral analysis are performed on the university VAX cluster using FORTRAN and IMSL (International Mathematical and Statistical Libraries) routines.

8. DESCRIPTION OF COLLECTED DATA

A significant amount of wind and building surface pressure data has been collected so far, in three acquisition modes, M01, M04 and M15. In each of the modes, record duration is 15 minutes. A description of each acquisition mode is given below.

M01 - This mode consists of pressures on the building's transverse section and meteorological data. Two hundred and forty-six records were collected from January to June 1989. This mode contains data from 11 Validyne transducers (see Table 1 for tap locations), 4 three-cup anemometers at 3, 13, 70 and 160 ft levels, 2 wind vanes at 3 and 160 ft levels, and temperature, barometric pressure and relative humidity at 13 ft level. All transducers and anemometry were sampled at 10 Hz; other instruments were sampled at 1 Hz. This data has been published previously [2].

M04 - This mode consists of pressures on the roof corner and meteorological data. Two hundred and eighty-seven records were collected from July, 1989 to May, 1990, 119 of which have stationary wind speeds and directions. This mode contains data from 11 Validyne transducers, 9 located on the roof corner and 2 at midwall taps (see Table 1 for tap locations). Meteorological data collected and sampling rates are the same as for mode M01 except that the wind vane at the 3 ft level was moved to 13 ft. This data is currently available and some of it has been published [7].

M15 - This mode consists of pressures on the building transverse section, roof corner, and along one purlin, internal pressure and meteorological data. This is the current acquisition mode. As of this writing, 127 records have been collected since April 1991, of which 34 are stationary. This mode contains data from 17 Validyne and 28 OMEGA pressure transducers (see Table 1 for tap locations), and all meteorological instruments. All transducers are sampled at 40 Hz, anemometry is sampled at 10 Hz, and other instruments are sampled at 1 Hz. Validyne transducers are analog low-pass filtered at 10 Hz, and OMEGA at 8 Hz. This data is being validated and will be analyzed in the near future.

Recorded data will be made available to all interested researchers. Summary statistics for all records in mode M04 are available on floppy disk, along with custom database software to assist in analysis of the data. Time histories are also available on request. Mode M15 data will be made available once it has been validated. For researchers using DOS-based personal computers, data can easily be sent on any size floppy disk, 20 MB Bernoulli removable cartridge (8-in.) or 600 MB erasable optical cartridge (5 1/4-in.) in ISO or Ricoh format. Data can also be sent via magnetic tape for use with mainframe computers.

In addition to the data, a package of information giving more details about the field laboratory is in preparation. This package will include complete instrument specifications, a topographical map, and numerous photographs of the surrounding terrain and other useful information.

9. REFERENCES

1 M.L. Levitan, K.C. Mehta, C.V. Chok, and D.L. Millsaps. An Overview of Texas Tech's Wind Engineering Field Research Laboratory, presented at the Sixth U.S. Conference on Wind Engineering (Houston, Texas), Journal of Wind Engineering and Industrial Aerodynamics, Vol. 36, pp. 1037-1046, Oct., 1990.

2 M.L. Levitan, J.D. Holmes, K.C. Mehta, and W.P. Vann. Field Measured Pressures on the Texas Tech Building, presented at the Eighth Colloquium on Industrial Aerodynamics and Wind Engineering (Aachen, W. Germany), accepted for publication in the Journal of Wind Engineering and Industrial Aerodynamics, 1989.

3 M.L. Levitan and K.C. Mehta. Texas Tech Field Experiments for Wind Loads Part II: Meteorological Instrumentation and Terrain Parameters, presented at the Eighth International Conference on Wind Engineering, London, Ontario, July 8-12, 1991.

4 D. Surry. Pressure Measurements on the Texas Tech Building-II: Wind Tunnel Measurements and Comparison with Full Scale, presented at the Eighth Colloquium on Industrial Aerodynamics and Wind Engineering (Aachen, W. Germany), accepted for publication in the Journal of Wind Engineering and Industrial Aerodynamics, 1989.

5 J.S. Bendat and A.G. Piersol. Random Data, Analysis and Measurement Procedures, 2nd ed., John Wiley and Sons, 1986.

6 M.L. Levitan. Statistical Analysis to Validate Full Scale Wind and Structural Response Data, master's thesis submitted to the Dept. of Civil Engineering, Texas Tech University, May, 1988.

7 K.C. Mehta, M.L. Levitan, R.E. Iverson, and J.R. McDonald. Roof Corner Pressures Measured in the Field on a Low Building, presented at the Eighth International Conference on Wind Engineering, London, Ontario, July 8-12, 1991.